

BIOCHEMISTRY & MOLECULAR BIOLOGY

Mechanisms of Gene Expression Regulation

A Molecular Paradigm Across Prokaryotic and Eukaryotic Systems

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The Regulatory Hierarchy

Gene expression refers to the multistep process that results in production of functional gene product, either RNA or protein. It is the basis for cellular adaptation, metabolic efficiency, and tissue differentiation across all domains of life.

Primary Levels of Control



Epigenetic & Transcriptional

Regulates whether genes are **accessible** and controls **when RNA polymerase starts copying DNA into RNA.**



Post-Transcriptional

Controls **what happens to mRNA after it is made**, including editing it (splicing), moving it out of the nucleus, and breaking it down when it is no longer needed.



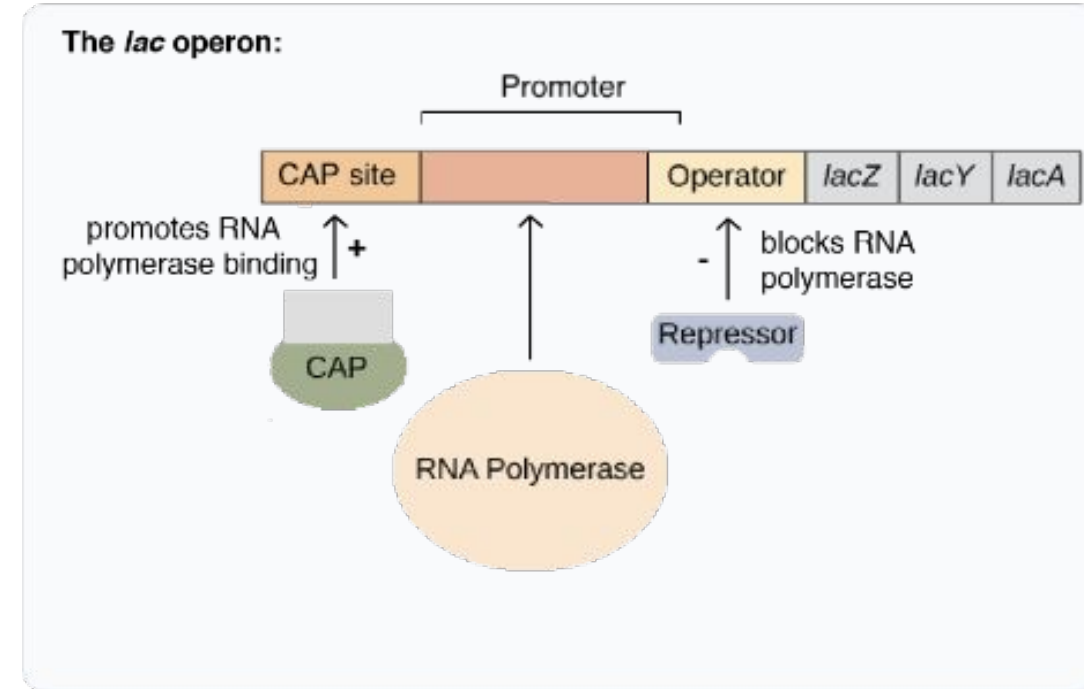
Translational & Protein

Regulates **protein production, protein modification, and protein breakdown.**

Prokaryotic Operon Model

Transcriptional Logic in Bacteria

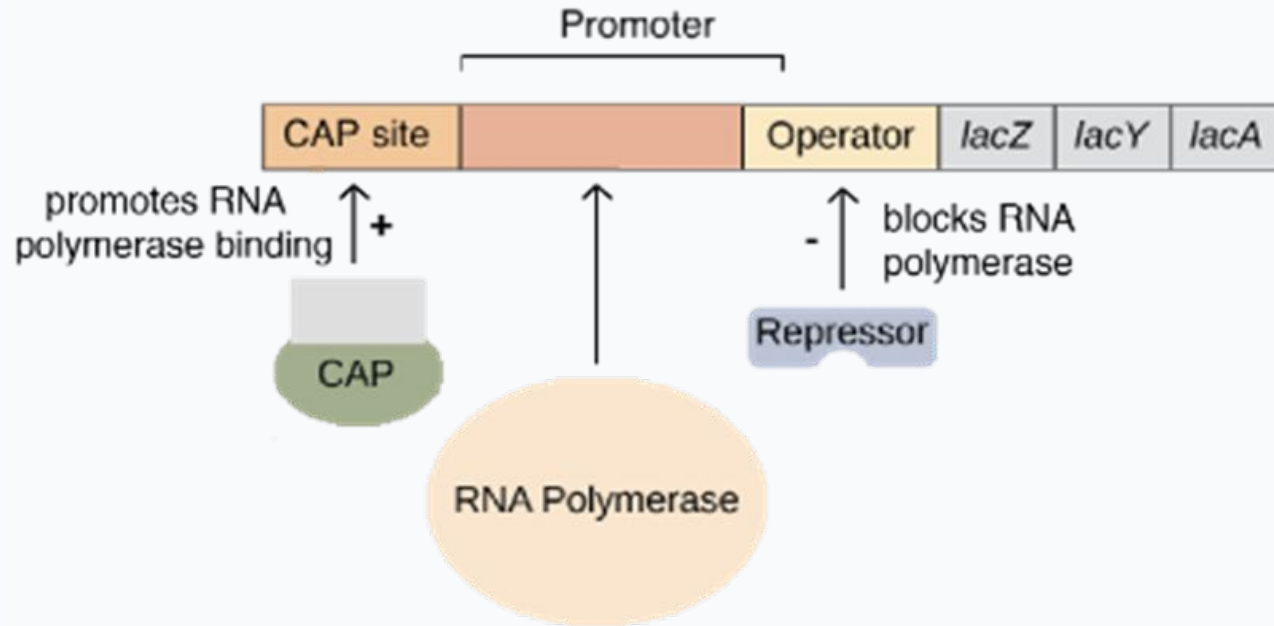
- 1. Operon Structure:** A group of **related genes** is controlled together and copied into **one mRNA molecule** (single polycistronic mRNA).
- 2. Negative Inducible (Lac Operon):** When **lactose is absent**, the **LacI repressor** blocks the genes, so they are **OFF**.
When **lactose is present**, it forms **allolactose**, which removes the repressor, turning the genes **ON**.
- 3. Catabolite Repression:** When **glucose is low**, **cAMP levels increase**. cAMP binds to **CAP** (catabolite activator protein), and together they help **turn transcription ON**.
- 4. Negative Repressible (Trp Operon):** When **tryptophan is low**, the genes are **ON** to make more tryptophan.
When **tryptophan is high**, it acts as a **co-repressor**, turning the genes **OFF**.



Easy way to remember:

- **Lac operon = Lactose turns genes ON.**
- **Trp operon = Tryptophan turns genes OFF.**
- **Low glucose = More cAMP-CAP = Stronger transcription.**

The *lac* operon:



Lactose Operon

Lac operon contains gene that codes for 3 proteins involved in the catabolism of disaccharide lactose.

lacZ gene codes for β -galactosidase which hydrolyzes lactose to galactose and glucose.

lacY gene codes for a permease which facilitates the movement of lactose into the cell.

lacA gene codes for thiogalactoside transacetylase which acetylates lactose.

Epigenetic & Chromatin Dynamics

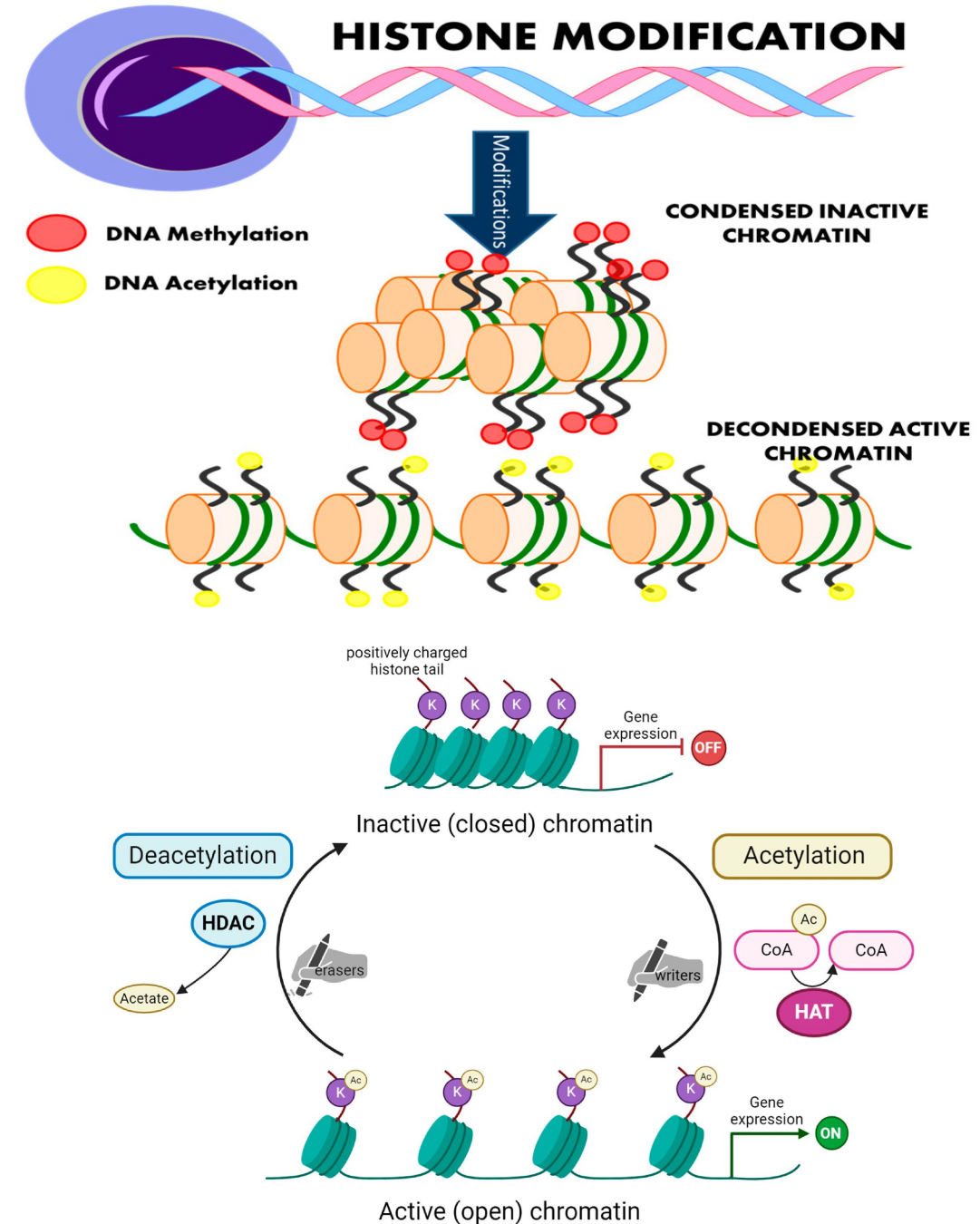
Histone Modifications

1.Acetylation (Gene ON):

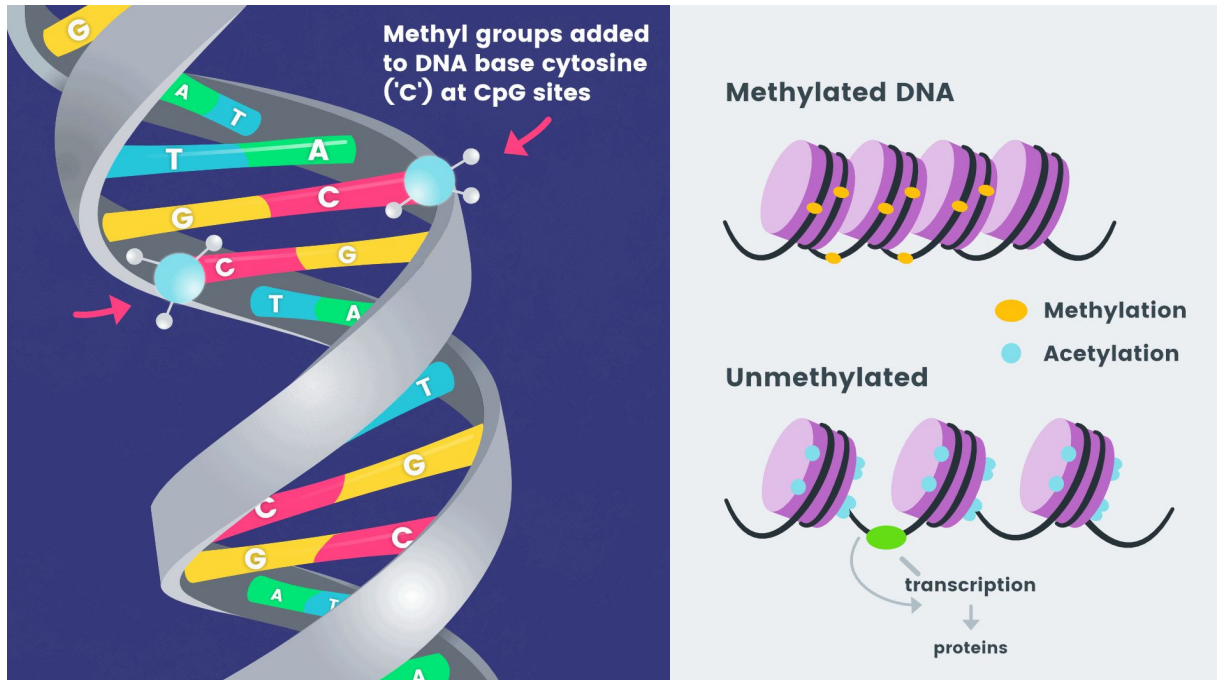
- Histone Acetyltransferases **HAT** enzymes add acetyl groups to histones neutralize positive lysine charges.
- This **loosens (open chromatin; euchromatin) the DNA**, making it easier for genes to be read(transcription).

2.Deacetylation (Gene OFF):

- Histone Deacetylases **HDAC** enzymes remove acetyl groups from histones restore charge.
- This **tightens (condensing chromatin; heterochromatin) the DNA**, making it harder for genes to be read(silence gene).



Epigenetic & Chromatin Dynamics



DNA Methylation

1.CpG Islands:

- **DNA Methyltransferases (DNMTs)** add **methyl (CH₃) groups** to **cytosine bases** in the promoter region of DNA.
- This process is called **DNA methylation**.

2.Gene Silencing:

- The methyl groups attract **methyl-CpG-binding proteins (MBDs)**, which block transcription factors from binding to DNA.
- As a result, transcription cannot start, so the gene is turned OFF.

Easy way to remember:

- DNMT = Adds methyl groups
- Methylated DNA = Transcription factors can't bind
- DNA methylation = Gene OFF (silenced)

Eukaryotic Transcription

Combinatorial Control Mechanisms

Starting Transcription:

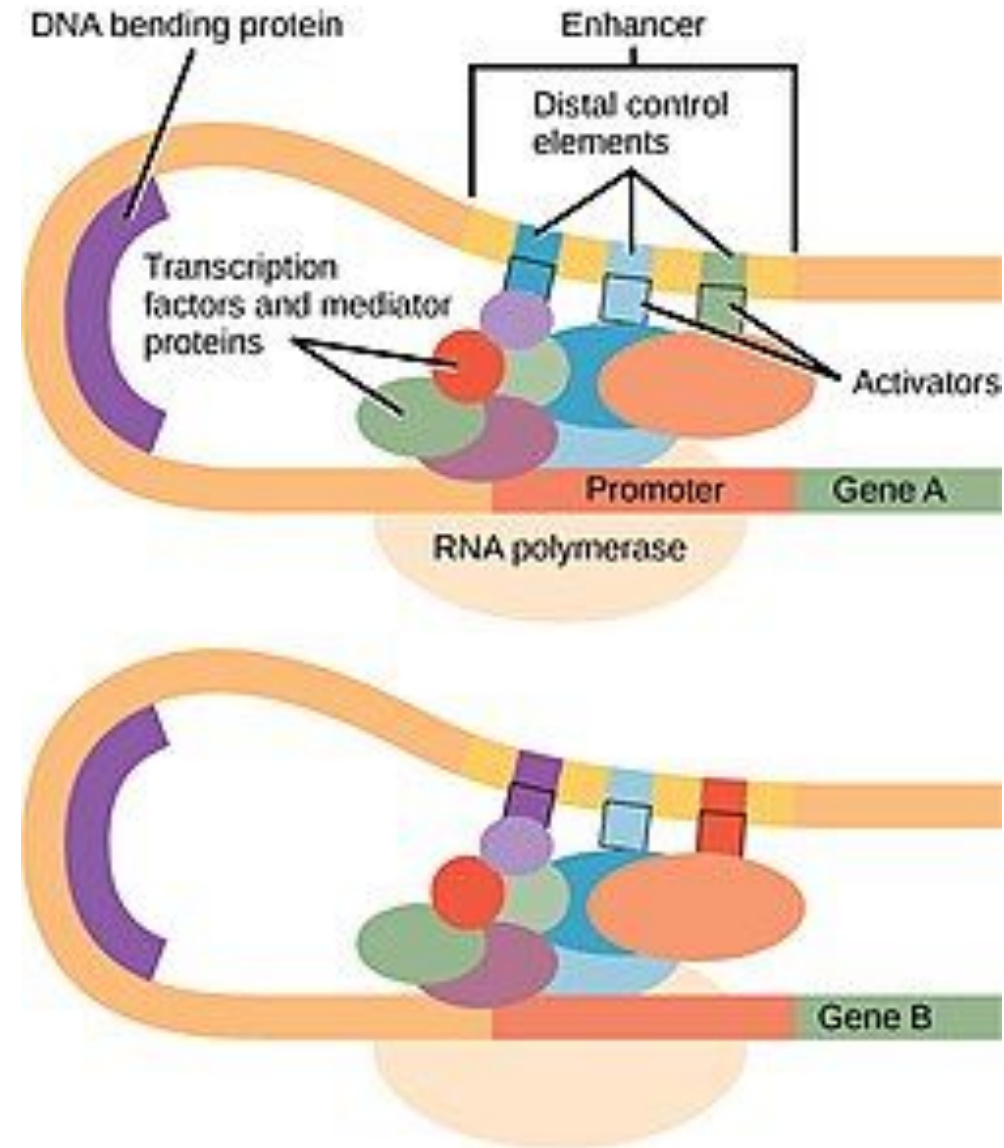
1. Eukaryotic **RNA Polymerase II** cannot start transcription by itself.
2. It needs **general transcription factors** (like **TFIID** and **TFIIB**) to bind to the **TATA box**(basal promoter elements) in the promoter.
3. Together, they form the **initiation complex**, which starts transcription.

Distal Regulatory Elements(Enhancers and Silencers):

1. **Enhancers** increase gene expression.
2. **Silencers** decrease gene expression.
3. They work by binding trans-acting **activators** or **repressors** proteins.

DNA Looping:

1. DNA bends into a loop so that **enhancers** can come close to the **promoter**.
2. The **Mediator Complex** helps connect them, making transcription easier.



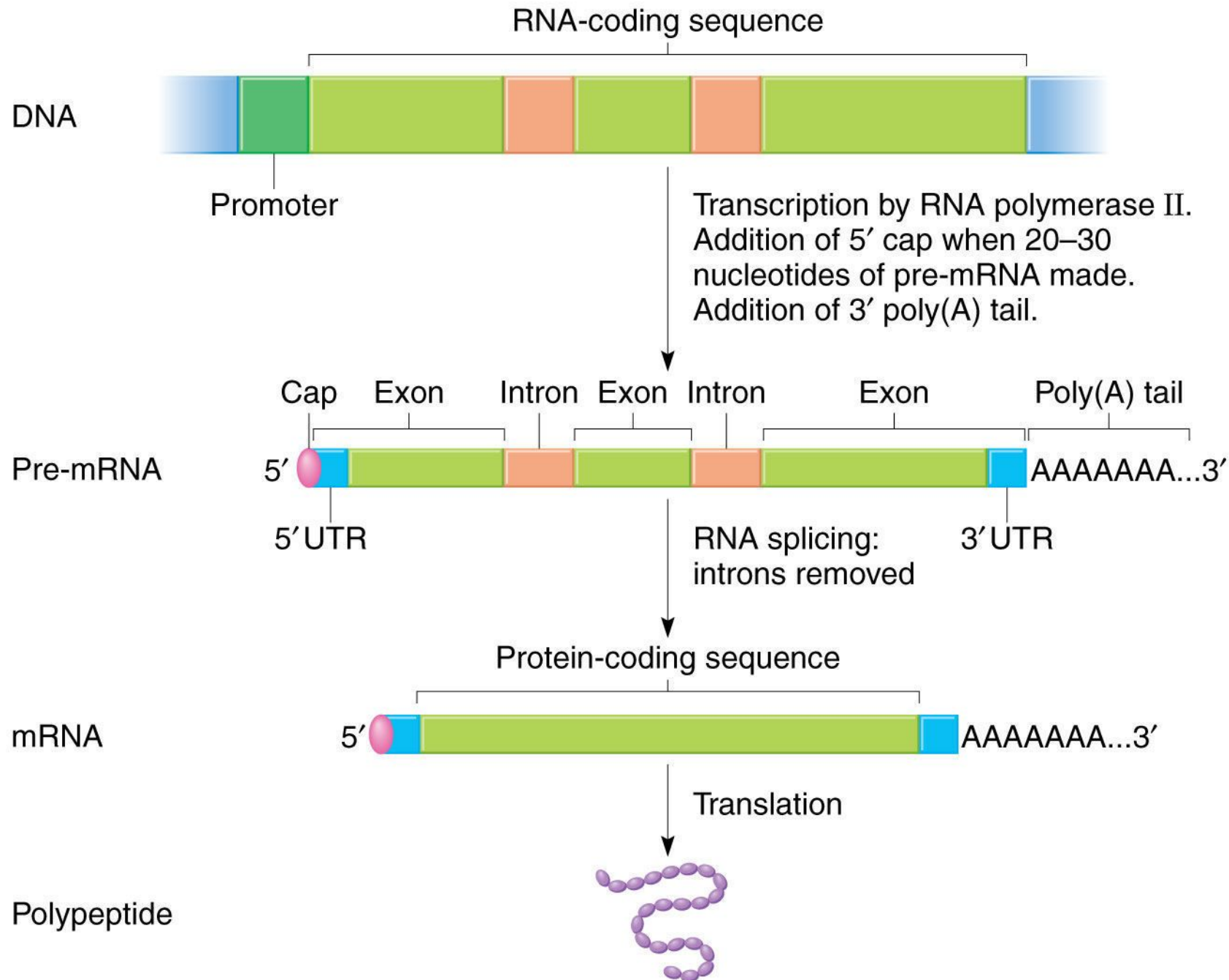
Post-Transcriptional Processing

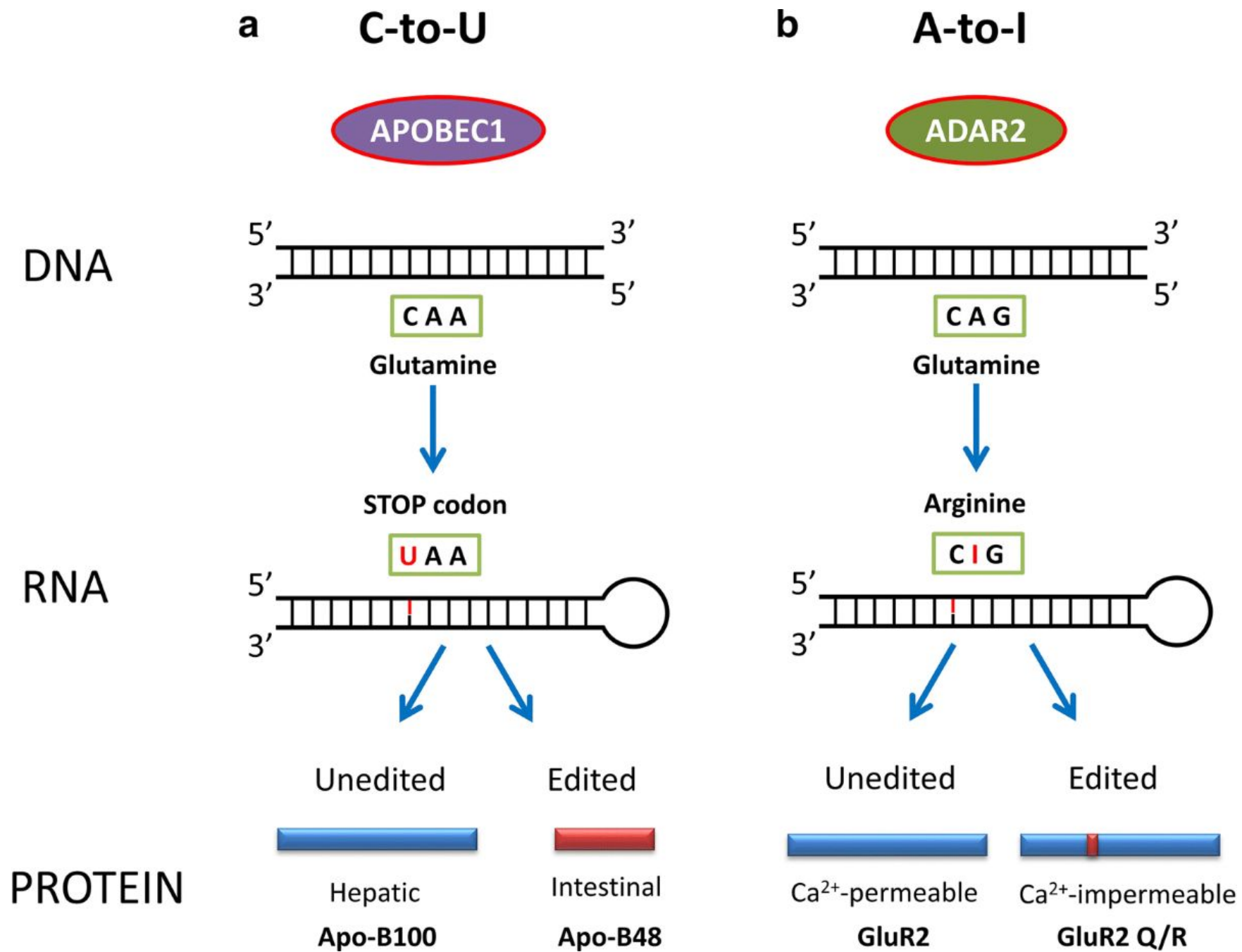
✓ **5' Capping & Polyadenylation:** Addition of 7-methylguanosine cap and 3' Poly-A tail protect the mRNA from damage and help it leave the nucleus.

✂ **Alternative Splicing:** Spliceosomes selectively removes some parts (introns) and joins the coding parts (exons). Different combinations of exons can make different protein isoforms from the same gene.

✎ **RNA Editing:** Enzymes like ADAR (A-to-I) and APOBEC (C-to-U) change some RNA bases after transcription. This can change the protein that is produced.

🕒 **mRNA Decay Dynamics:** AU-rich elements (AREs) in 3' UTRs and RNA-binding proteins control how long mRNA lasts before it is broken down.
Longer-lasting mRNA = More protein produced.
Shorter-lasting mRNA = Less protein produced.





MicroRNA Gene Silencing

~2

Nucleotides in Functional

miRNA

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RNA Interference (RNAi) Pathway

MicroRNAs (miRNAs):

- miRNAs are small non-coding RNAs that help control gene expression.
- They are processed by Drosha in the nucleus and Dicer in the cytoplasm.

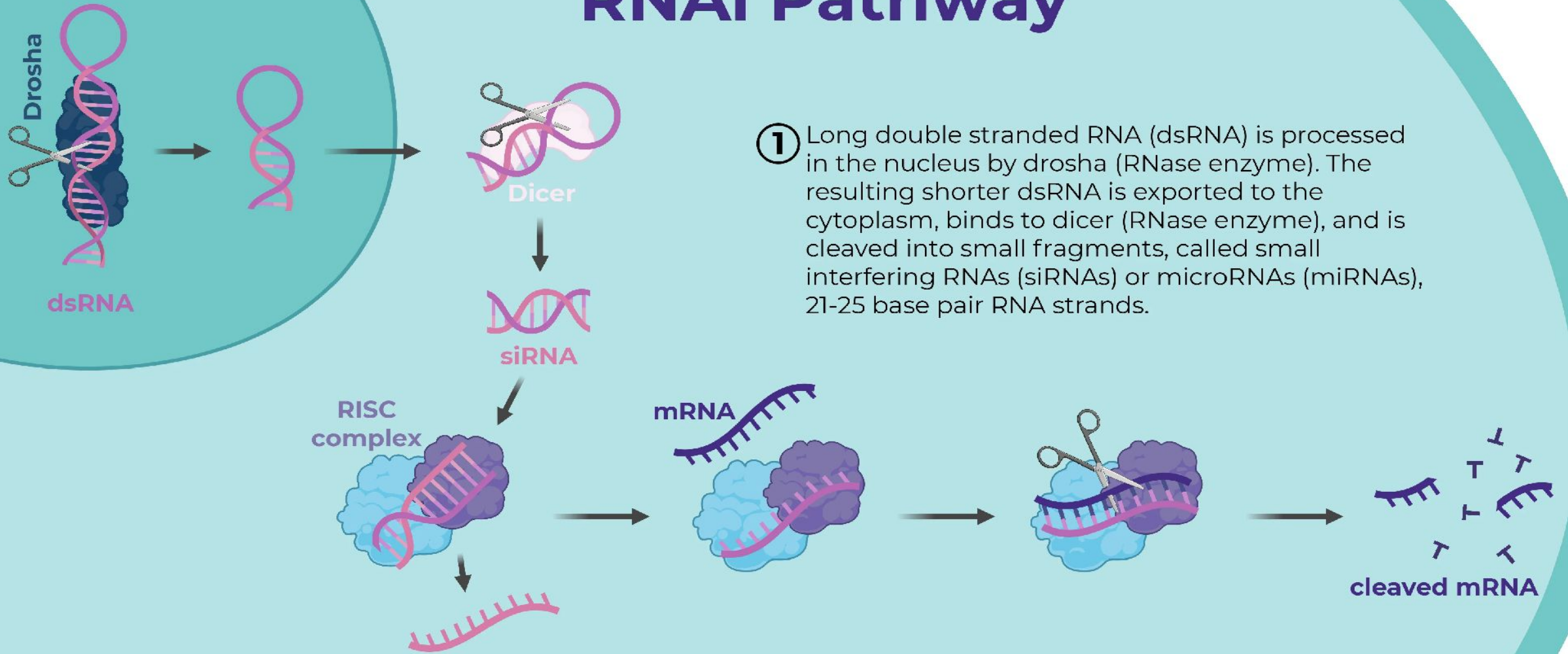
RISC Complex:

- The mature miRNA joins the RISC (RNA-induced Silencing Complex)
- .RISC uses the miRNA to find and bind the matching target mRNA.

Gene Silencing:

- When miRNA binds to the 3' UTR of mRNA, it:
Blocks translation (less protein is made), or
Breaks down the mRNA, so no protein is produced.

RNAi Pathway



① Long double stranded RNA (dsRNA) is processed in the nucleus by drosha (RNase enzyme). The resulting shorter dsRNA is exported to the cytoplasm, binds to dicer (RNase enzyme), and is cleaved into small fragments, called small interfering RNAs (siRNAs) or microRNAs (miRNAs), 21-25 base pair RNA strands.

② The siRNA or miRNA is recruited by the RISC complex, unwinds, and incorporates into a protein-RNA complex.

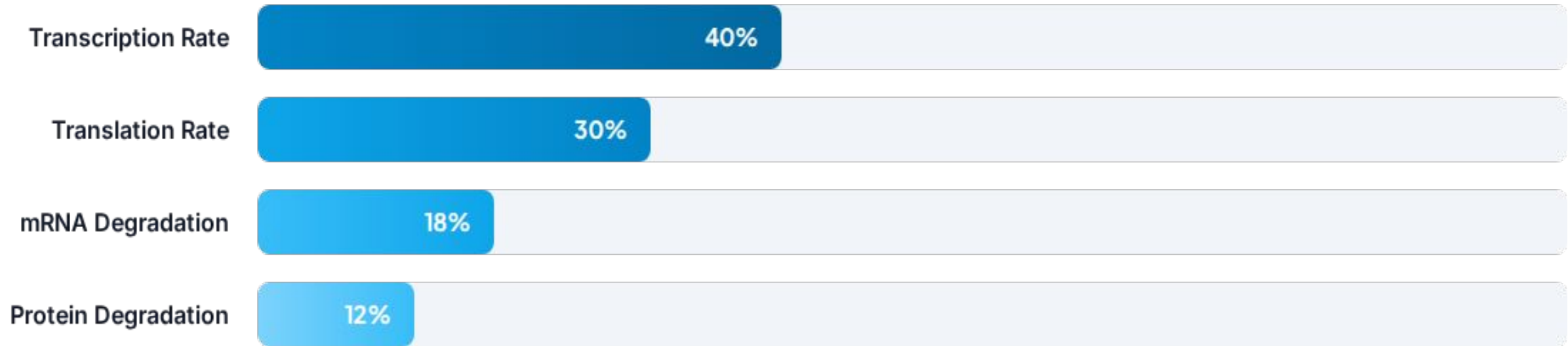
③ RISC + siRNA or miRNA bind to a complementary, targeted messenger RNA (mRNA).

④ mRNA is cleaved in a specific site and then degraded in the cell, thereby disrupting protein synthesis of the target gene.

Prokaryotes vs Eukaryotes

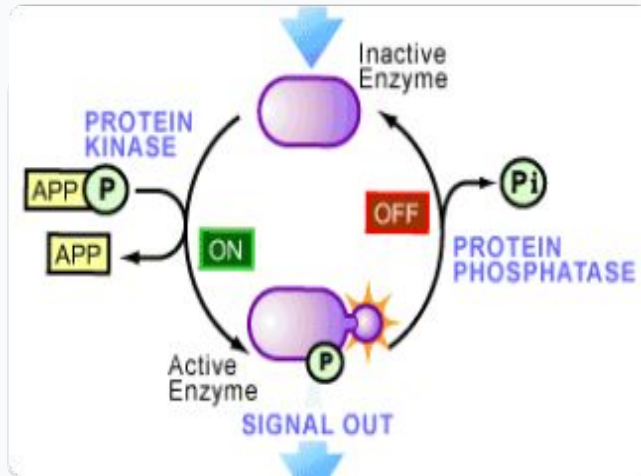
Feature	Prokaryotic Regulation	Eukaryotic Regulation
Cellular Compartment	Coupled transcription & translation (Cytoplasm)	Spatially segregated (Nucleus & Cytoplasm)
Gene Arrangement	Polycistronic operons with shared promoters	Monocistronic genes with complex enhancers
Chromatin Structure	Naked DNA / NAP packaging (No nucleosomes)	Histones, nucleosomes, euchromatin/heterochromatin
mRNA Processing	Minimal or none; immediate translation	5' Cap, 3' Poly-A tail, introns spliced out
Primary Control Site	Transcription Initiation	Multiple levels (Epigenetic to PTMs)

Contributions to Protein Levels



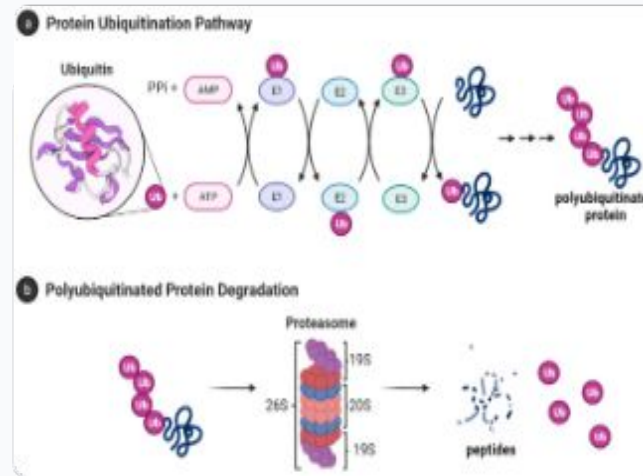
Biochemical quantitative modeling indicates that transcription and translation rates jointly account for ~70% of variation in steady-state cellular protein concentration.

Post-Translational Modifications



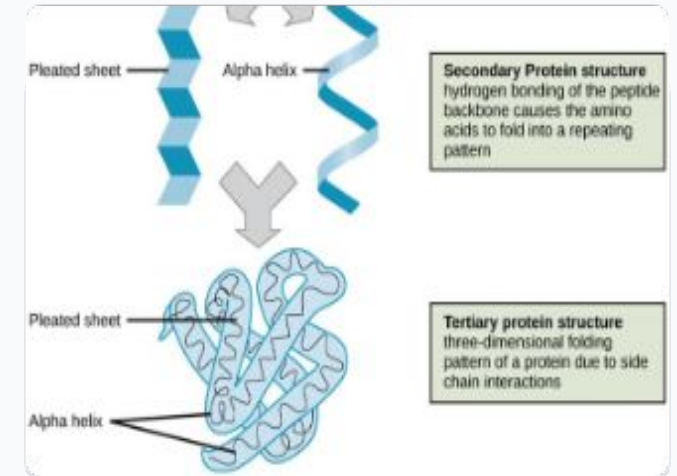
Phosphorylation

Kinase enzymes add a phosphate group to Ser/Thr/Tyr residues protein. This turns the protein ON or OFF and helps send signals inside the cell.



Ubiquitination

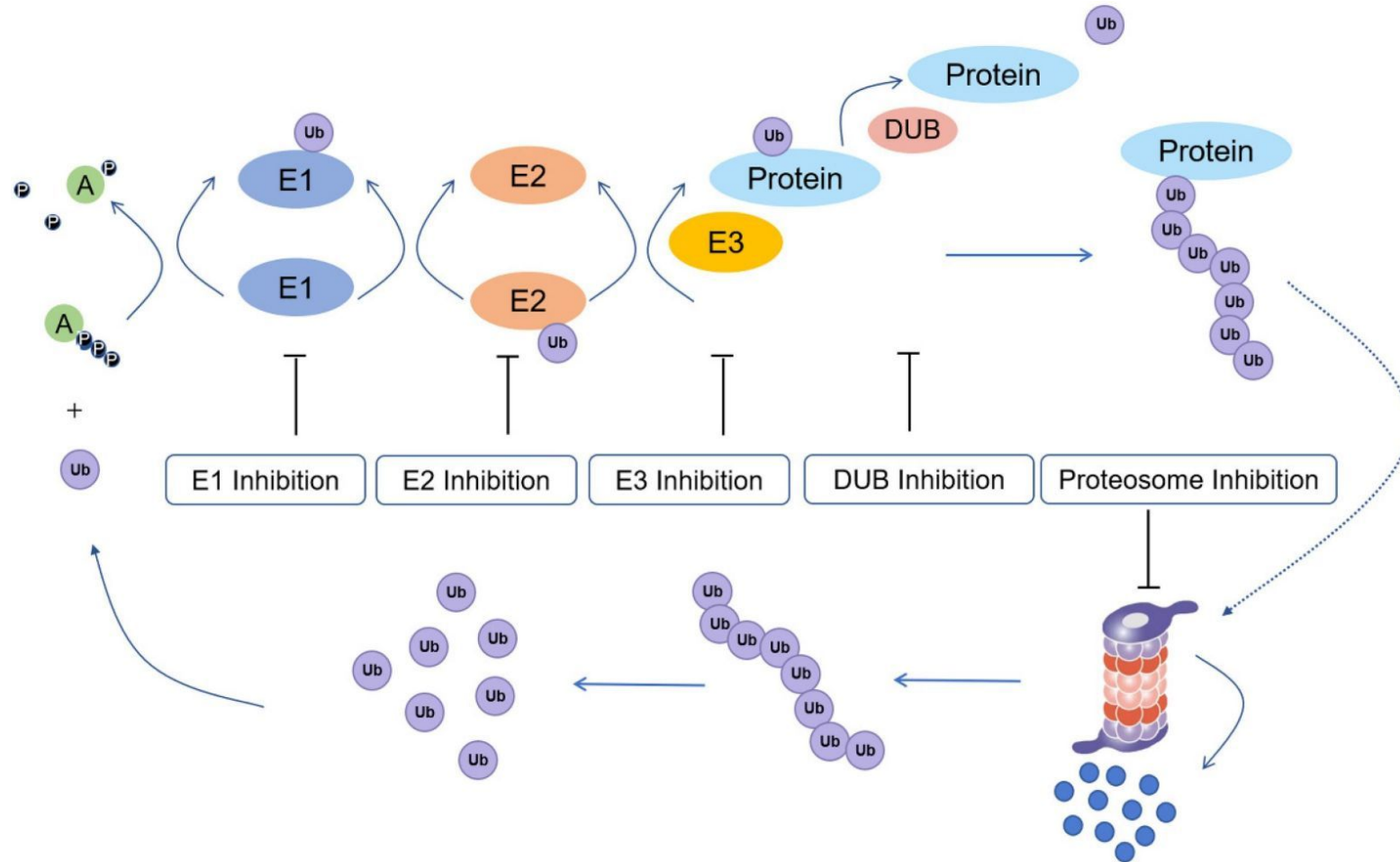
E1, E2, and E3 enzymes attach ubiquitin to unwanted or damaged proteins. The 26S proteasome recognizes the ubiquitin tag and breaks down the protein.

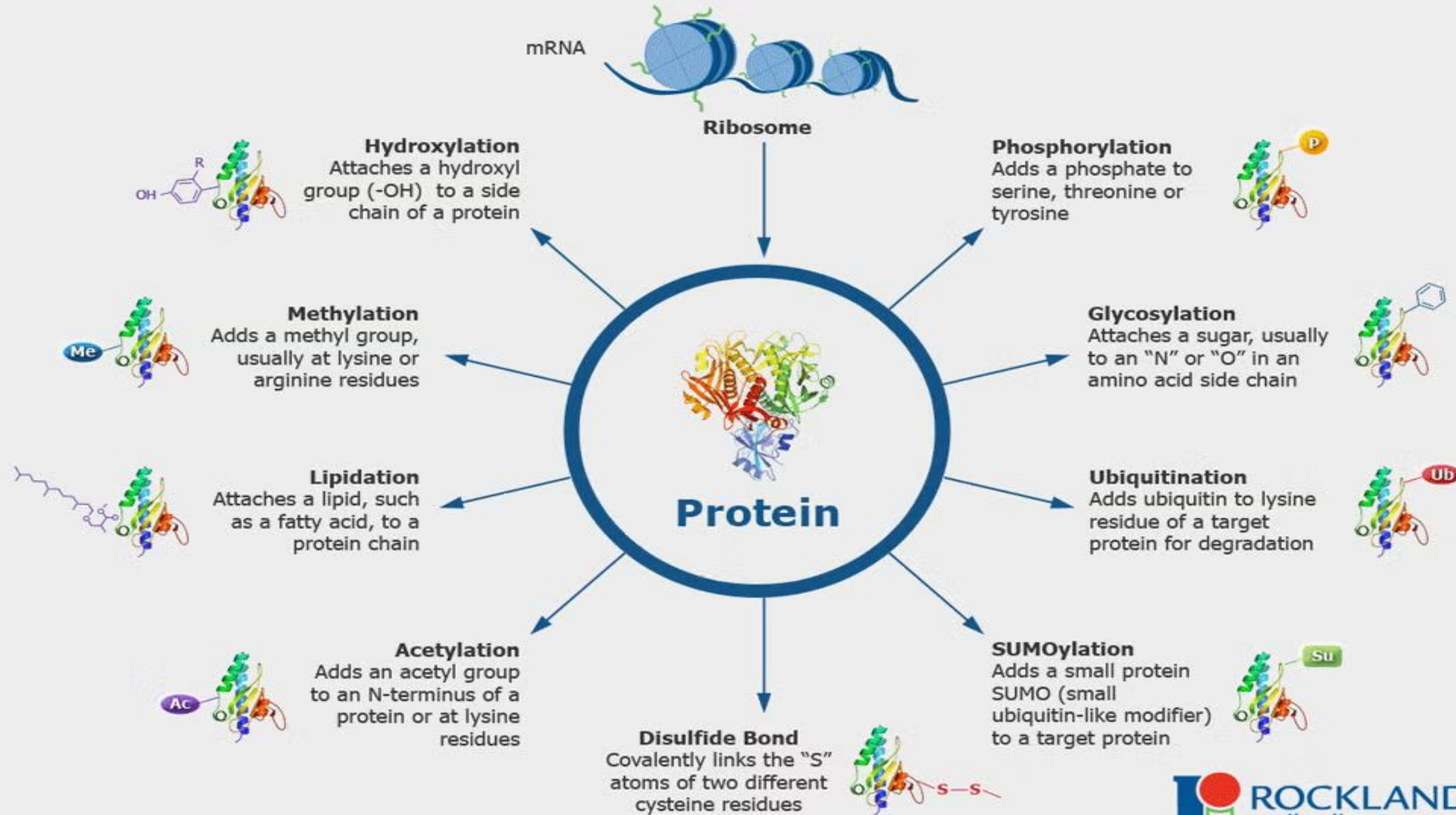


Chaperones

Heat Shock Proteins (HSPs) help proteins fold into the correct shape (tertiary folding). They also prevent proteins from clumping together (aggregation).

UBIQUITINATION





WATCH VIDEOS!

Protein Synthesis: Transcription and Translation

<https://youtu.be/oefAl2x2CQM>

Gene Expression and Regulation

<https://youtu.be/ebIpkw3XapE?si=0VNCSYAa5LeDi7vL>

Questions & Discussion

Biochemistry of Gene Expression Regulation